AI-Powered Math Tutoring Chabot

Teaching Systems of Equations Through Guided AI Interaction

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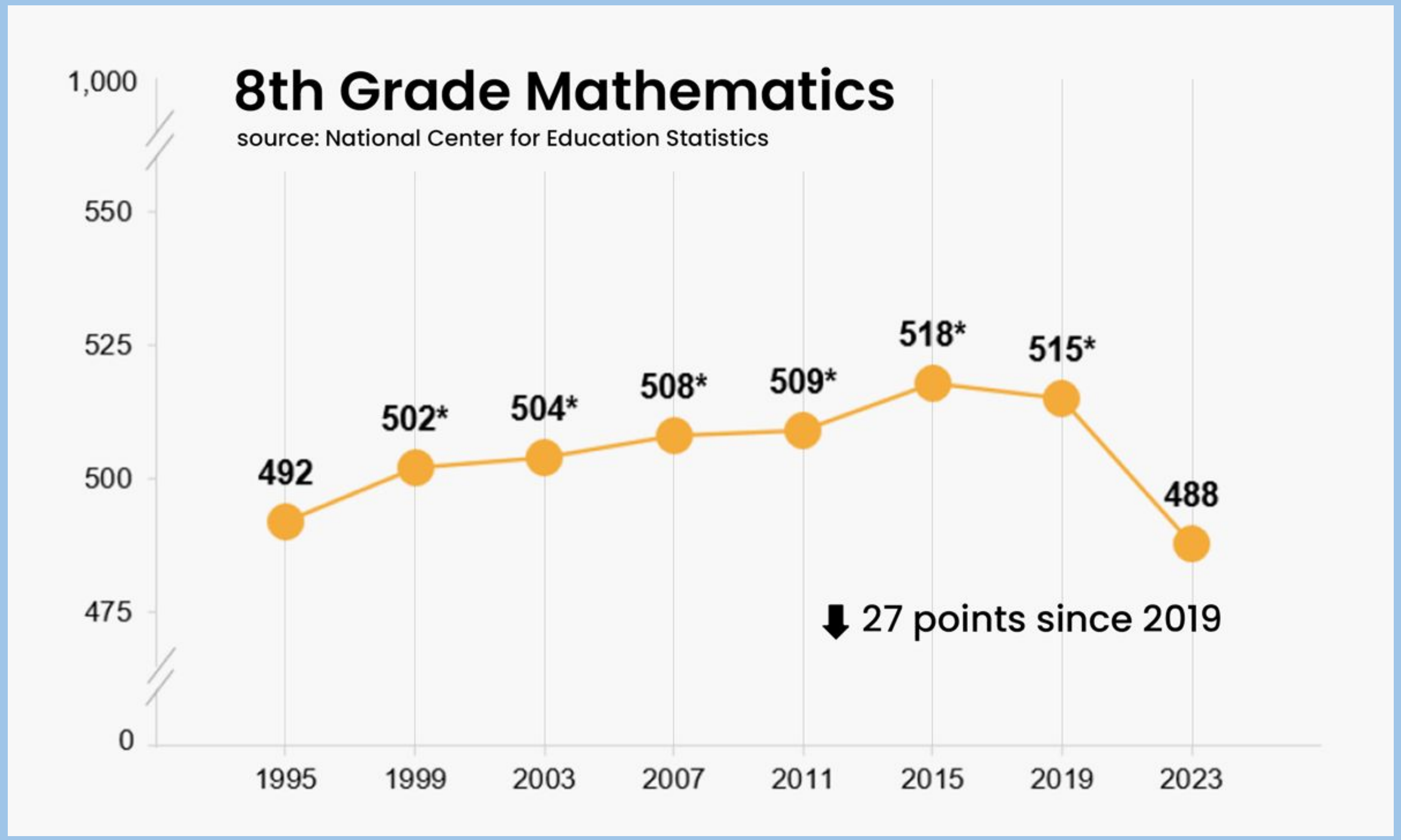
The Problem:

Why did we build this?



Math Scores -

US Math performance has declined significantly since COVID-19

Increased Prevalence of LLMs

Could that help the problem?

Teacher workload -

Large class sizes (Average US class size = 25 students)

1 teacher cannot help everyone at once

Guidance from Our Tutor would be

- Flexible and On demand
  - Available 24/7, including outsides school hours
- Low Pressure

Our Solution

Practice Mode

- Chatbot Generates a Problem
- Student works through it
  - with step by step AI guidance should the student need it.

Chat Mode

- Student drives conversation
  - Student can
    - Ask questions
    - paste own problem
    - Get guided help

Problem Types

Linear	Quadratic	Exponential	Systems

Difficulty Levels

Easy - solve given equations	Medium - word problems w/ equations provided	Hard - word problems given, student must figure out equations to model given word problem

Now For A Quick Demo...

Learning Theory: Schema

How we help students build pattern recognition

Schema (def) - organizing knowledge in memory through experience - allows experts to solve problems and choose strategies efficiently

- I always think of chess, minesweeper, solving a rubik’s cube
  - people who can do these activities well and quickly use patterns to understand situations

Category Scaffolding

- User always knows problem type
  - Could help them spot patterns

Examples and Non Examples

- Students will categorize problems
  - Linear, Quadratic, exponential systems
  - System with 1 solution, 2 solutions, infinitely many solutions, no solutions.

Pattern Recognition

- Students internalize what problems look like

Learning Theory: Constructivism

Learners build knowledge like a tower/pyramid

Each layer rests upon previous layers

You can't skip layers

All of our content is f fundamental to later math

Hard mode: Derive equations from word problem

Medium mode : Word problem + Given Equations

Easy mode: solve systems of given equations

Prior knowledge: lines can intersect at 0, 1, or infinitely many points

Exponential Systems (adv)

Quadratic Systems (Intermediate)

Linear System (Foundational)

Progression By Problem Type

System Architecture

Frontend UI

- Mode Selection, Question Interface

HTTPS

Backend Server (Flask)

Session Manager

- Tracks active module + difficulty.
- Saves Progress + Answers
- In Memory conversation history
- Discarded on page regresh

Problem Config

Content + Assessment

- Generates problem by type + level
- Streams hints + explanation
- Calls LLM for Socratic Guidance
- Calls SymPy to verify steps

Prompt History

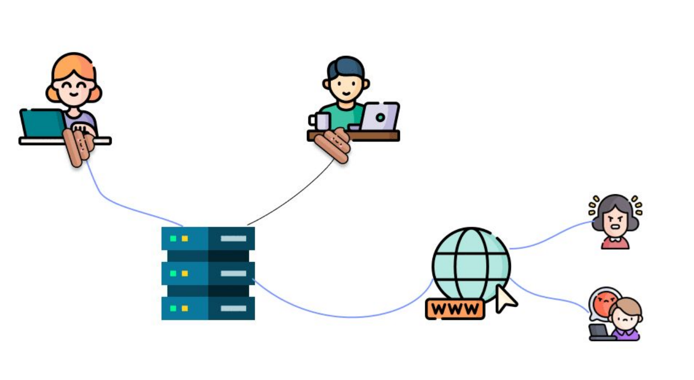
Student Step

Local LLM (gemma3:12b via Ollama)

- Generates questions + hints
- Encourages student
- Does NOT do arithmetic

SymPy (math verifier)

- Validates student steps algebraically
- Checks final answers about solution set
- LLM cannot override SymPy verdicts



Conversation History:

- Full message thread
- Cleared on page refresh

Session State

- Current problem
- Cleared on Page Refesh

Prompt Optimization

Structured A/B testing across three prompt variants

Variant A
Baseline

- minimal prompt optimization
- light restriction against giving answers immediately

Avg: 3.53

Variant B
Hardened

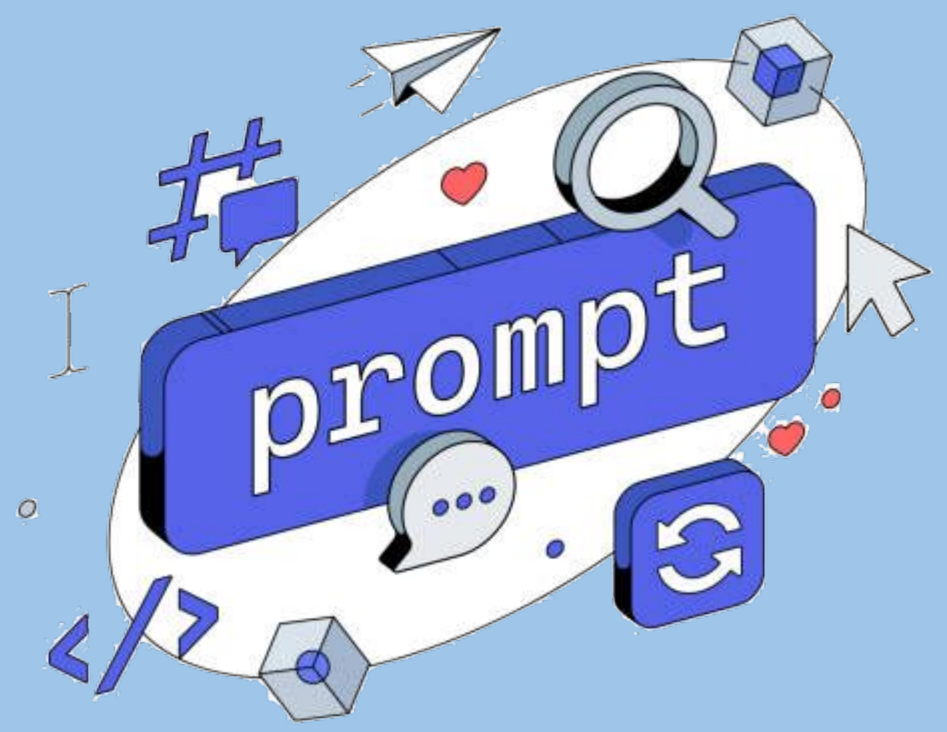
- Added Jailbreak resistance
- Friendlier tone
- No SymPy acknowledgement
- no backtracking

Avg: 3.53

Variant C
Scaffolded

- Example responses
- Fixed LaTeX delimiters
  - (\(...) vs $...$)
- more elaborate step-gating

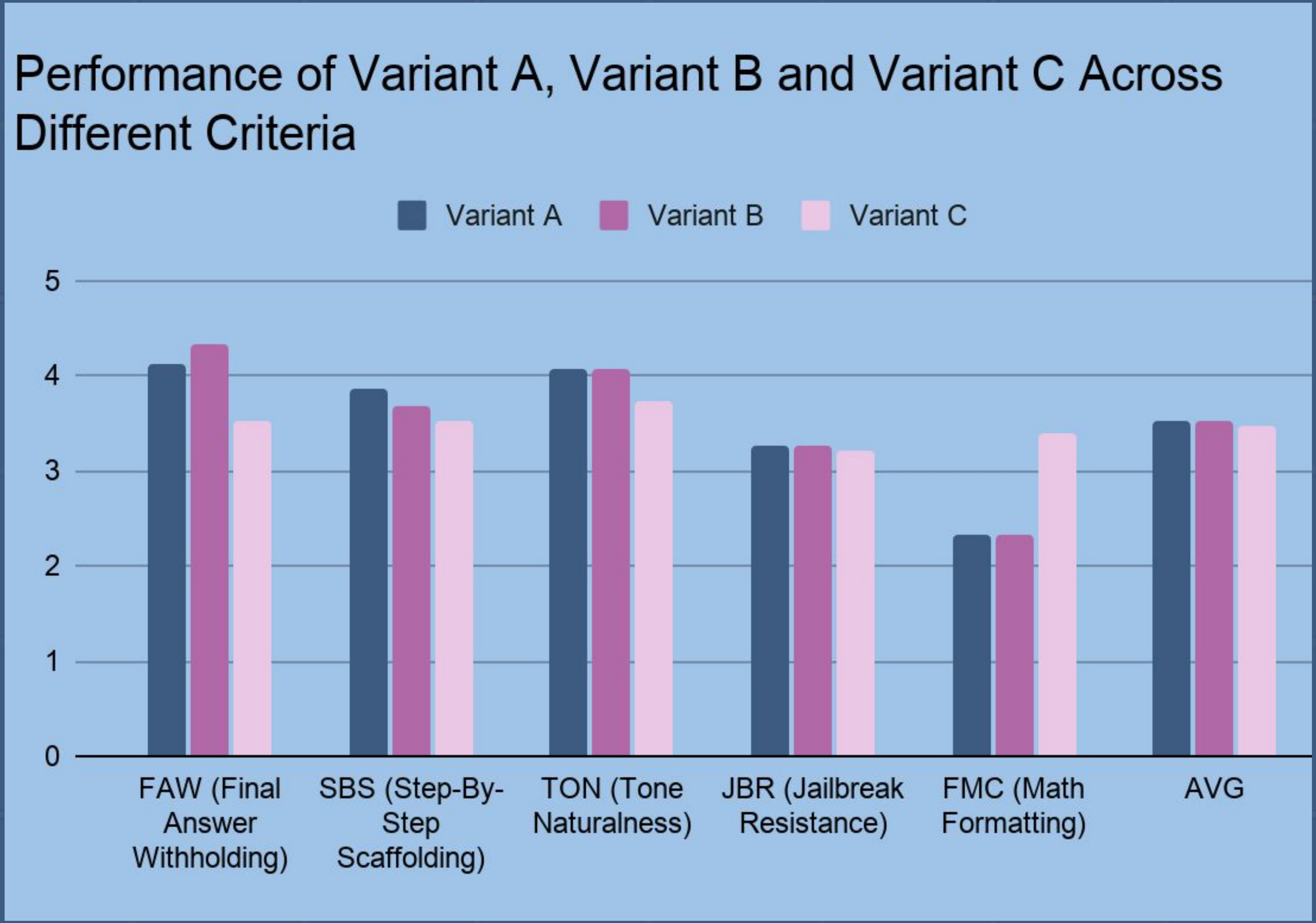
Avg: 3.48



5 Evaluation Criteria (scored 1-5 by LLM-as-judge across 15 test Scenarios

FAW: Final Answer Withholding	SBS: Step-by-Step Scaffolding	TON: Tone Naturalness	JBR: Jailbreak Resistant	MFC: Math Formatting

Why We Choose Variant C

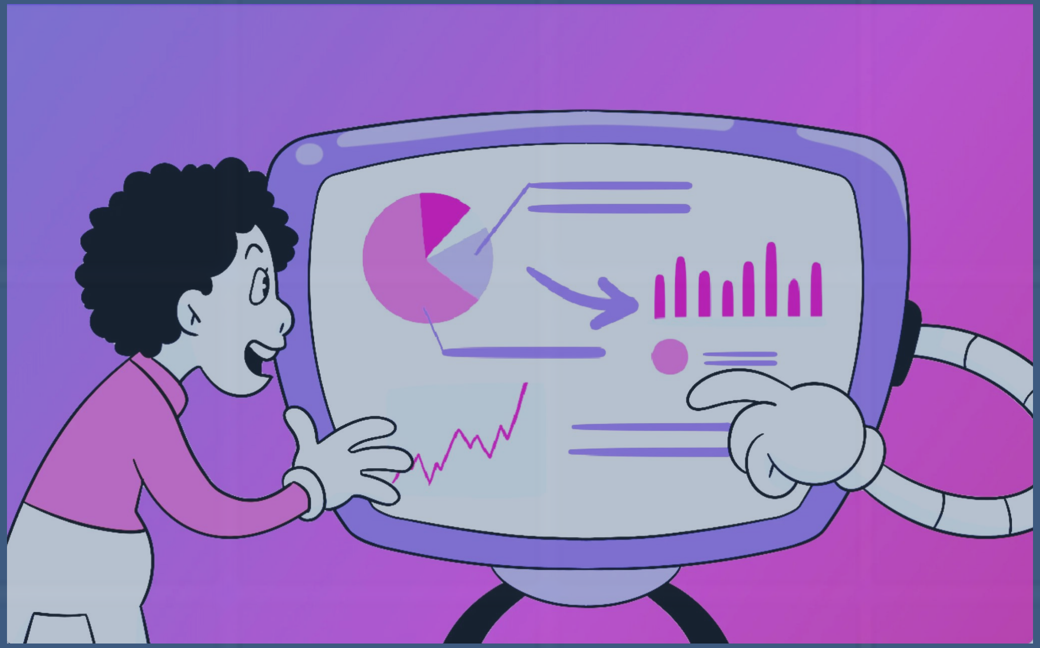


MFC: 2.33 → 3.40
Very important
Variant C = easiest way to optimize

FAW = Fixable with Instructions

Scores are underestimated.
Variant C's True score is higher.

Jailbreak is model-intrinsic



User Testing

5 Participants,
4 College Students,
1 Math Educator

5/5 Rated Response speed “just right’	**⅘** Rated interface system ⅘ or 5/5	**⅘** said it would be more helpful than google or youtube	**0** found the tone condescending

What Worked

- Real-time Q&A on specific problems
- Encouraging, non-condescending tone
- 60% of respondents said the tuor explained the “why”, not just the “how”
- Low pressure environment

What didn’t

- One participant received an incorrect answer
- Broken LaTeX rendering in some responses
- One participant jailbroke the bot (Lucy’s contrarian friend)

Ethical Considerations

Our users = students → perhaps more vulnerable

Supplement, not replacement

Tool cannot replace a teacher student relationship

Equity and access

- Local Ollama setup favors well resourced households/schools.
- Deployment must prioritize access
  - Perhaps a lightweight, browser based interface

Safety with Minors

- One user bypassed safeguards
- Production leads app-layer protections beyond prompt alone

Mathematical Accuracy

- SymPy handles arithmetic, LLM can make conceptual errors

Data Privacy

- Currently stores nothing
- Deployment must comply with FERPA/COPPA

The Socratic Model

- Withholding answers for autonomy, teaching effectiveness.

Productionisation Plan: Hosting

Current Limitations

- Requires local GPU
  - Not Cloud Deployable
  - SSE streaming breaks serverless
    - 29s API Gateway
    - Lambda and similar services won't sustain session

Recommended: Fly.io + Bedrock

- Fly.io : Docker Containers, Native SSE support
- Auto scale-to-zero during off school hours
- AWS Bedrock for LLM interface
  - Keeps data in-account

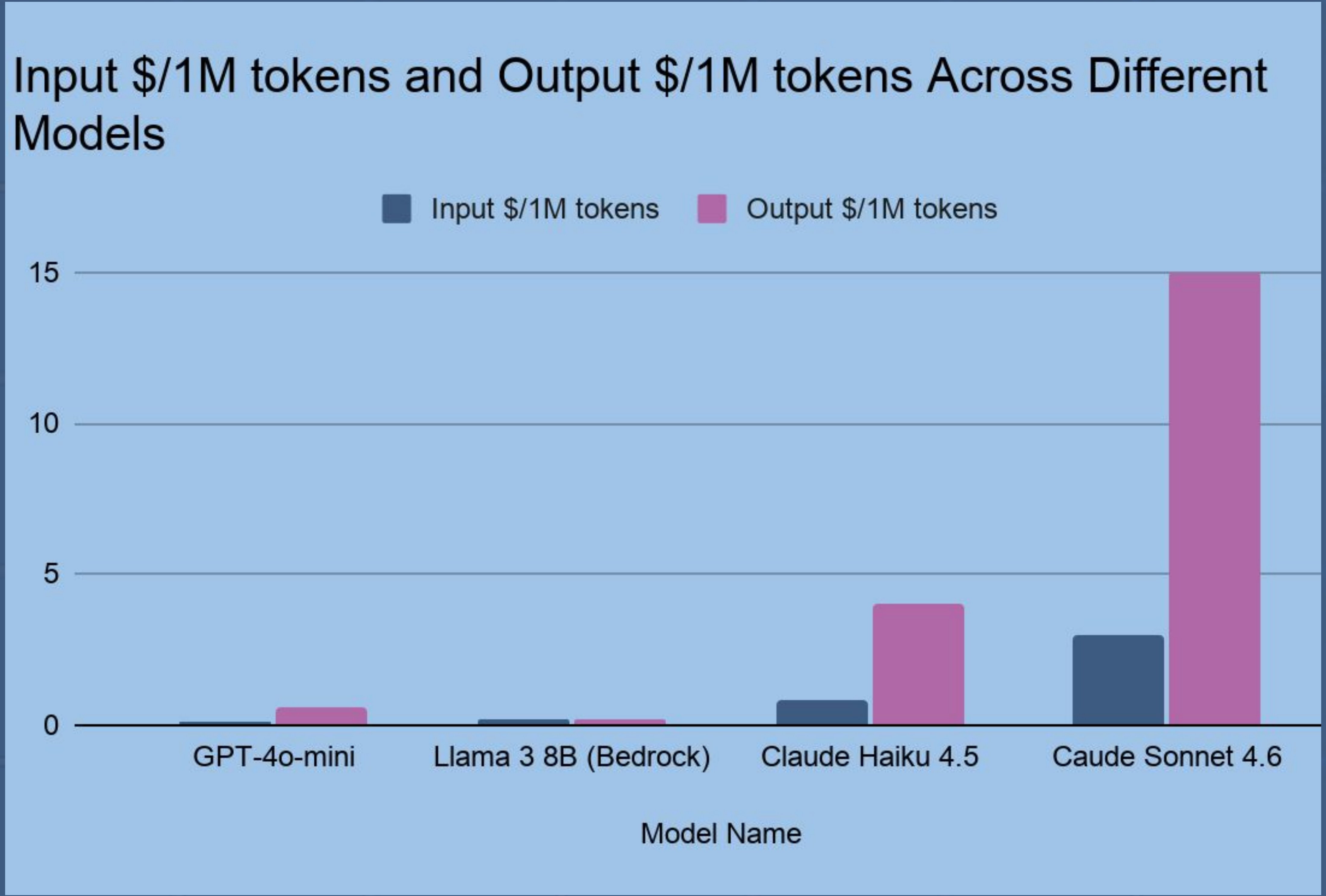
Component	Service	Purpose
App Server	Fly.io	SSE Streaming
Session Store	Upstash Redis	Conversation Hitory
Static assets	Cloudflare CDN	JSS, CSS KaTeX
LLM inference	Bedrock/Open AI	Replace Ollama

Productionisation Plan: Cost

LLM’s Job = teach.

LLM’s job ≠ math

Math handled by SymPy



- Input = messages that get sent to the model (user’s messages)
- Output = messages from the model
- This stuff costs money for whoever is running the app
  - Input = cheaper
    - reading is always cheaper than writing

Recommendation:

General ed-tech

GPT-4o

Low cost, fastest

K-12 (COPPA)

Bedrock Llama 38B

Data stays with AWS

Avoid For Now

Claude Sonnet / GPT-4o

10-20x cost with minimal pedagogical gain

Failure Modes and Privacy

What Can Go Wrong -
And how we'd address it

API Outage

- Circuit breaker after 2 failures / 30s
- Friendly UI fallback message
- Cache static hints for common steps
- Secondary provider fallover

Wrong Math Answer

- SymPy already validates every student step
- Contradiction rate tracked per model version
- Post generation SymPy verification pass planned

Harmful Output/Jailbreak

- LLama Guard (Bedrock) or OpenAI Moderation API
- Rate limiting: 60 messages / hour / session
- Flag repeated violations to instructor
- Applicant layer safeguards, not just prompt

K-12 Data Privacy: FERPA, COPPA, SOPIPA

AWS Bedrock = only clearly compliant path

Future Directions

- Increase scope of math topics
- More focus on interactivity for graphing problems
- Try and move away from basic chatbot UI

Thank you for listening!

Any Questions?